Norma vectorială și norma matricială

(1) Norma vectorială

$$\|x\| : \mathbb{R}^n \longrightarrow \mathbb{R}_+$$

Prop

$$\|x\| = 0 \iff x = 0 \qquad \|x\| > 0$$
$$\|\alpha \cdot x\| = \alpha \|x\| \qquad \alpha \in \mathbb{R}$$
$$\|x + y\| \leq \|x\| + \|y\|$$

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}$$

$$p = 1 \Rightarrow \|x\|_1 = \sum_{i=1}^{n} |x_i|$$ Norma Manhattan

$$p = 2 \Rightarrow \|x\|_2 = \sqrt{\sum_{i=1} x_i^2}$$ Normă Euclidiană

$$\sqrt{\langle x, x \rangle} = \sqrt{x^t \cdot x}$$

$$p = \infty \Rightarrow \|x\|_\infty = \lim_{p \to \infty} \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p} =$$

$$= \max\{|x_i|\} \cdot \lim_{p \to \infty} \left| 1 + \sum_{i=1}^{n} \left(\frac{x_i}{\max\{|x_i|\}} \right)^p \right|$$

$$= \max\{|x_i|\}$$

(2) Norma matriciale

$$\|A\| : \mathbb{R}^{n\times m} \longrightarrow \mathbb{R}_+$$

Prop

$\|A\| > 0$ se $\|A\| = 0 \Rightarrow A = 0 \quad \forall A \in \mathbb{R}^{n\times m}$

$\|\alpha A\| = |\alpha| \cdot \|A\| \quad \forall A \in \mathbb{R}^{n\times m}$

$\|A + B\| \leq \|A\| + \|B\|$

consistenza: $\|A \cdot B\| \leq \|A\| \cdot \|B\|$

$$\|A\|_F = \sqrt{\sum_i \sum_j |A_{ij}|}$$

Norme subordinate

$$\|A\|_p = \sup_{\substack{x \in \mathbb{R}^n \\ x \neq 0}} \frac{\|A \cdot x\|_p}{\|x\|_p}$$

$$p = 1 \Rightarrow \|A\|_1 = \max_j \sum_{i=1}^{n} |A_{ij}|$$

$$p = 2 \Rightarrow \|A\|_2 = \sqrt{\rho(A \cdot A^t)} = \sigma_2(A) = \sqrt{\mathrm{Tr}(A^t \cdot A)}$$

$$\rho(A) = \max \{ |\lambda(A)| \}$$

$\rho(A)$ ↓ raggio spettrale

$|\lambda(A)|$ ↓ val. singolare

$$p = \infty \Rightarrow \|A\|_\infty = \max_i \sum_{j=1}^{n} A_{ij}$$

Condiționarea SEL (Sisteme de ec. liniare)

$Ax = b$

$\tilde{A}\tilde{x} = \tilde{b}$

$\tilde{A} = A + \delta A$

$\tilde{b} = b + \delta b$

$\tilde{x} = x + \delta x$

$$\varepsilon_x = \frac{\|\delta x\|}{\|x\|}$$

$$(A + \delta A)(x + \delta x) = b + \delta b$$

$$Ax + A\delta x + \delta A x + \delta A \delta x = b + \delta b$$

$$A\delta x + \delta A x + \delta A \delta x = \delta b$$

$\delta b \sim 0$

1°) $\delta A \delta x \simeq 0$

$$A\delta x + \delta A x = 0$$

$$A\delta x = -\delta A x \quad | \cdot A^{-1} \text{ la st.}$$

$$\delta x = -A^{-1}\delta A x$$

$$\|\delta x\| \leq \|A^{-1}\| \cdot \|\delta A\| \cdot \|x\| \quad \Big| : \|x\|$$

$$\underbrace{\frac{\|\delta x\|}{\|x\|}}_{\text{eroare relativă (eroare la ieșire)}} \leq \underbrace{\|A^{-1}\| \cdot \|A\|}_{\text{număr de condiționare}} \cdot \underbrace{\frac{\|\delta A\|}{\|A\|}}_{\text{eroarea la intrare}}$$

$$\boxed{\varepsilon_x \leq k(A) \cdot \varepsilon_A}$$

2°) $\delta A \delta x \neq 0$

$$A\delta x + \delta A x + \delta A \delta x = 0$$

$$A\delta x = -\delta A x - \delta A \delta x \quad | \cdot A^{-1} \text{ la st.}$$

$$\delta x = -A^{-1}\delta A x - A^{-1}\delta A \delta x$$

$$\|\delta x\| \leq \|A^{-1}\| \cdot \|\delta A\| \cdot \|x\| + \|A^{-1}\| \cdot \|\delta A\| \cdot \|\delta x\| \quad \Big| : \|x\|$$

$$\varepsilon_x \leq k(A) \cdot \varepsilon_A + k(A) \cdot \varepsilon_A \cdot \varepsilon_x$$

$$\varepsilon_x \left(1 - k(A) \cdot \varepsilon_A\right) \leq k(A) \cdot \varepsilon_A$$

$$\varepsilon_x \leq \frac{k(A) \cdot \varepsilon_A}{1 - k(A) \cdot \varepsilon_A} \simeq \varepsilon_A$$

dacă $k(A) \simeq 1$

Obs

Matricele ortogonale au $\kappa(A)=1$

Funcții Matlab norm(A) = norma 2 a lui A

norm(A,p) = norma p a lui A

cond(A,p) = numărul de condiționare al lui A

Exerciții

Calculați numărul de condiționare (folosind Matlab)

Hilbert
$$H_4=\begin{pmatrix} 1/2 & 1/3 & 1/4 & 1/5 \\ 1/3 & 1/4 & 1/5 & 1/6 \\ 1/4 & 1/5 & 1/6 & 1/7 \\ 1/5 & 1/6 & 1/7 & 1/8 \end{pmatrix}$$

Wilson
$$W=\begin{pmatrix} 5 & 7 & 6 & 5 \\ 7 & 10 & 8 & 7 \\ 6 & 8 & 10 & 9 \\ 5 & 7 & 9 & 10 \end{pmatrix}$$

$$R=\begin{pmatrix} 10 & 1 & 4 & 0 \\ 1 & 10 & 5 & -1 \\ 4 & 5 & 10 & 7 \\ 0 & -1 & 7 & 9 \end{pmatrix}$$